



Chapter 10

Dependable Systems

Reliability, safety, security, resilience — and the evidence needed to trust them.

[CRISIS](#)[MAP](#)[MECHANISM](#)[TRADE-OFF](#)[EVIDENCE](#)[VERDICT](#)

TIMEBOX: 30 sec

YOUR TASK Enter the lecture looking for one engineering decision, not a list of definitions.



RULE 01 · Crisis + decision + unknown

Engineering crisis: decide under uncertainty

Start with the failure scenario, not vocabulary.

A dependable-looking system can still fail people if the wrong property, layer, or evidence is ignored.



TIMEBOX: 60-90 sec

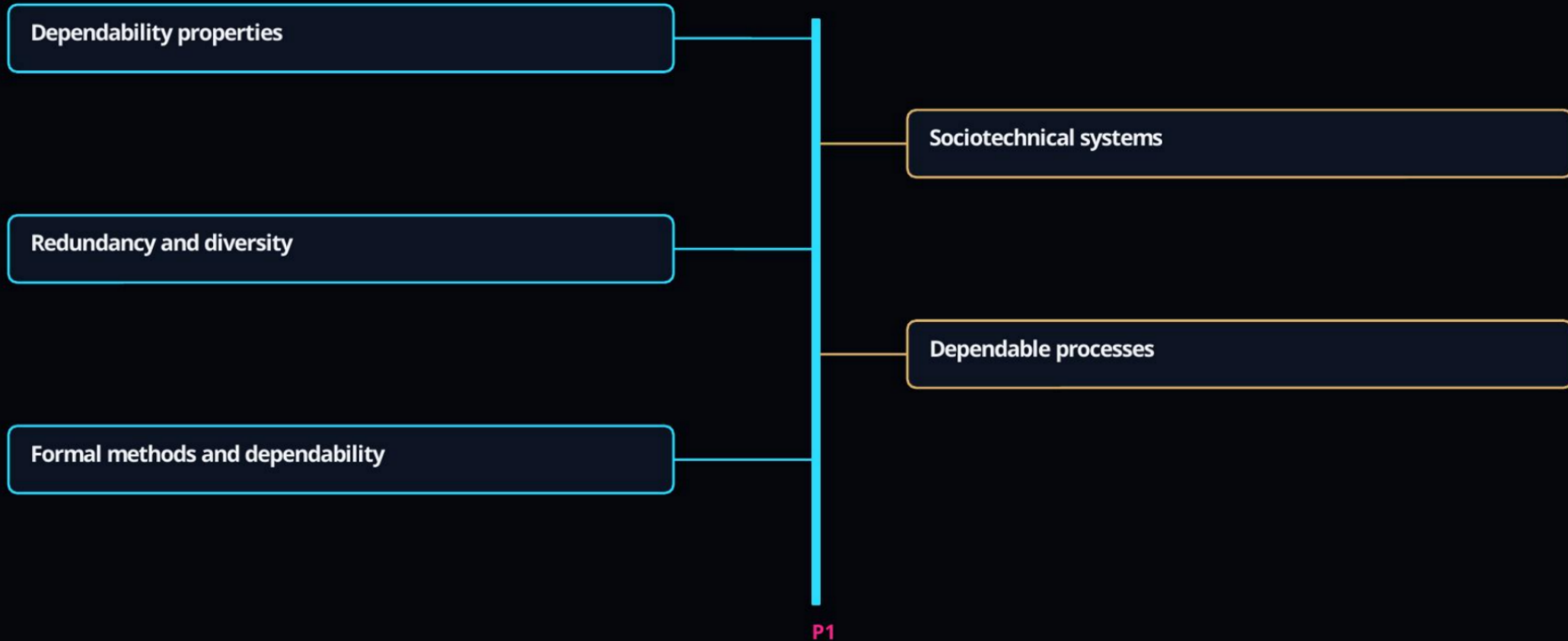
YOUR TASK State the crisis, the decision, and the unknown that could change the verdict.



RULE 02 · Domain spine

Domain spine

See the chapter as a map before reading details.



TIMEBOX: 60-90 sec

YOUR TASK Use the map to explain which source family controls the decision.



RULE 03 · Exactly five CLOs

Five learning outcomes

Every outcome requires visible evidence, not recognition alone.

- 1 Explain dependability as trust across reliability, availability, safety, security, and resilience.
- 2 Trace sociotechnical layers and failures.
- 3 Compare redundancy/diversity and process/formal assurance.
- 4 Build a bounded decision with evidence.
- 5 Defend the decision with counter-evidence and uncertainty.

TIMEBOX: 60-90 sec

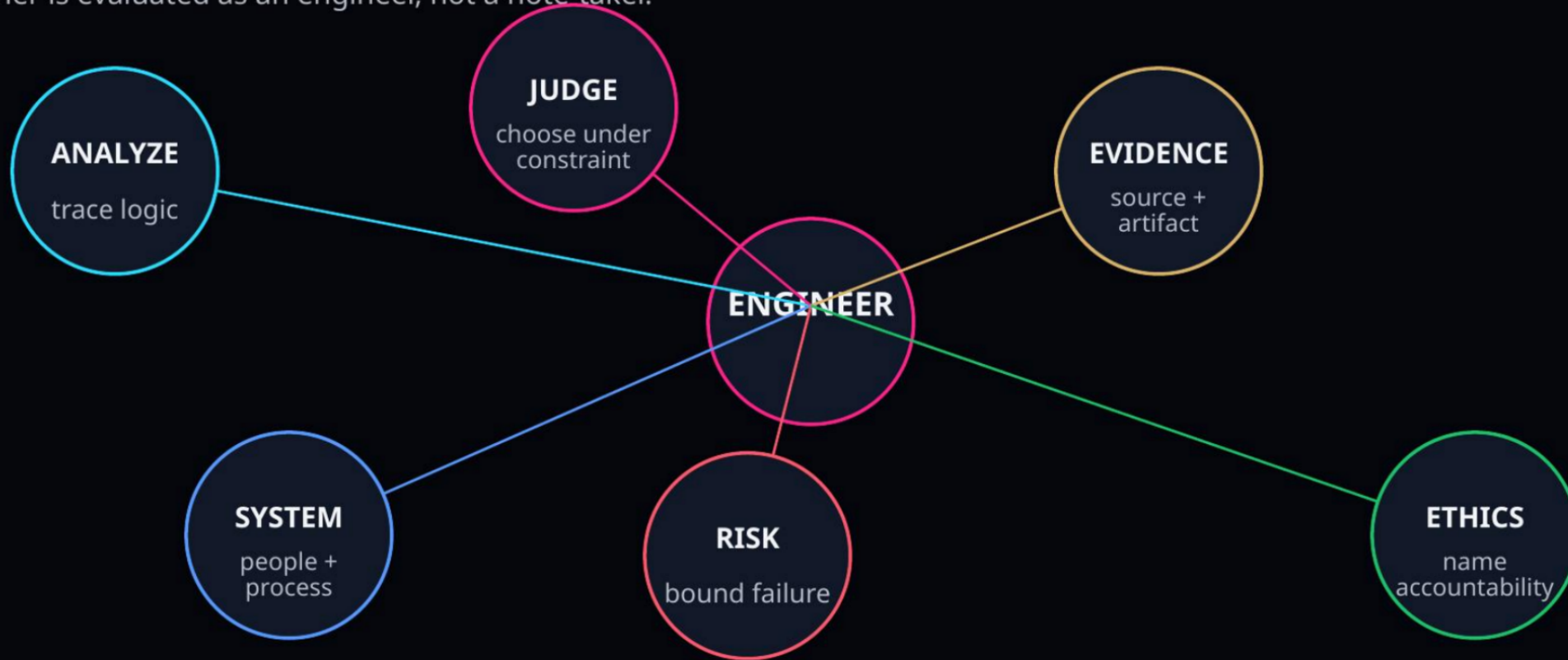
YOUR TASK Choose the outcome you expect to find hardest and explain why.



RULE 04 · Six engineering capabilities

Six engineering capabilities

The learner is evaluated as an engineer, not a note-taker.



TIMEBOX: 60-90 sec

YOUR TASK Identify which capability would fail first if the source evidence were misunderstood.



RULE 05 · Predict - constraint - derive - name

Prediction gate

Do the reasoning before naming the principle.

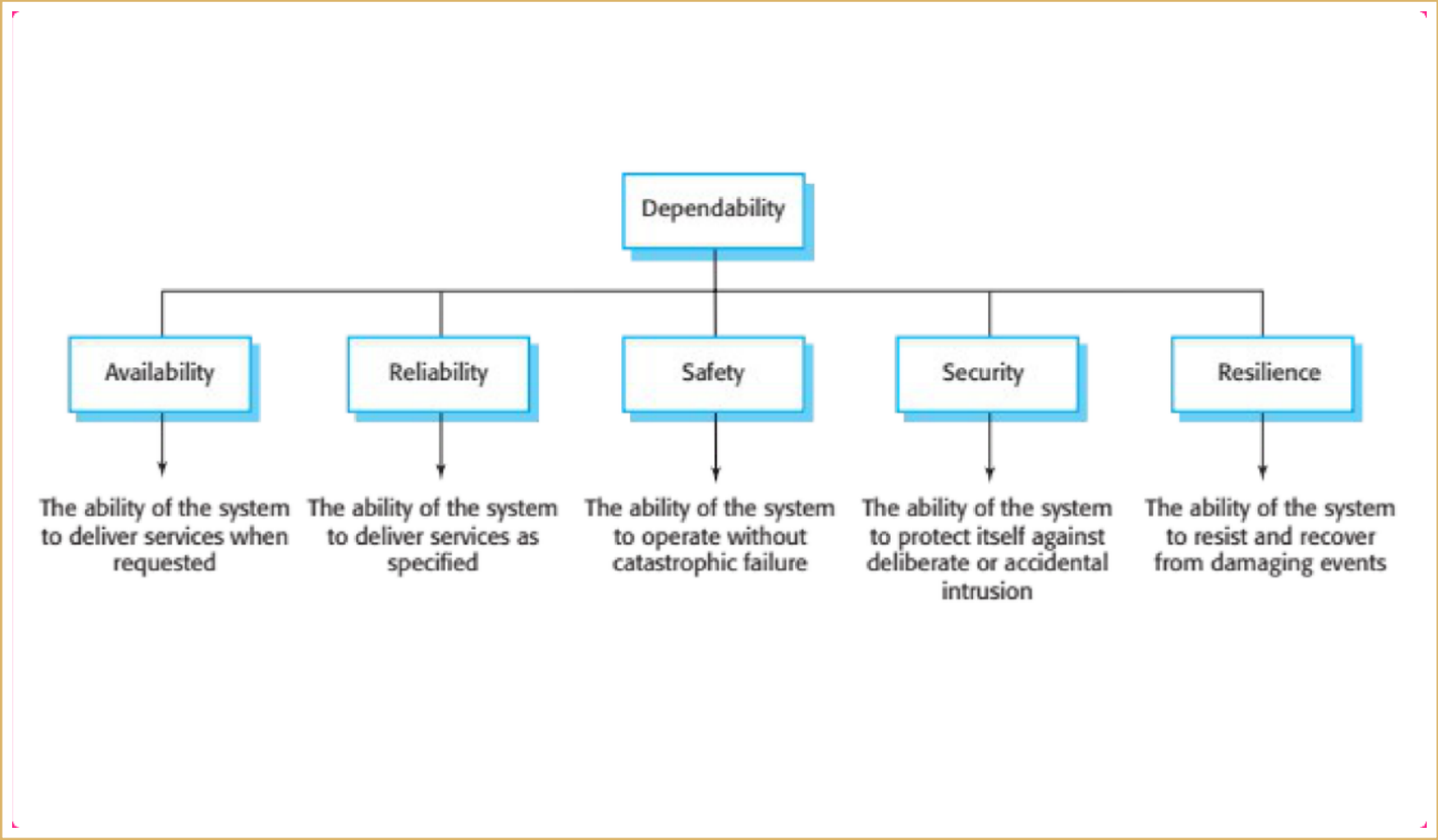
This sequence blocks post-hoc explanation and forces source-grounded reasoning.



TIMEBOX: 60-90 sec

YOUR TASK Commit to a prediction, constraint, derivation, and principle before seeing the worked answer.

Dependability properties



READ THE SOURCE FIGURE

- Dependability is not one property; it combines availability, reliability, safety, security, and

- Use the taxonomy to decide which promise the system must defend first.

Decision rule: shrink commentary before shrinking the evidence.



Dependability properties — precise definitions

Keep source definitions visible; the diagram is the map, not the whole lesson.

Core service properties

- Availability: service is ready when needed.
- Reliability: service continues correctly over time.
- Safety: operation avoids catastrophic harm.

Protection and continuity

- Security: service resists malicious compromise.
- Resilience: service continues after disruption and recovers from disturbance.

DECISION BRIDGE

Which property changes the deployment decision, and what evidence would prove the conflict?

TIMEBOX: 60-90 sec

YOUR TASK Pick one property that could conflict with another and give the deployment consequence.



RULE 07 · Structure + trace/apply

Dependability properties move together

Trace one property into another before applying the concept.



Dependability is not a single property; it is the interaction students must reason about.

TIMEBOX: 60-90 sec

YOUR TASK Start with one property, then trace what other property it threatens.



How dependability is actually achieved

Dependability is built before deployment and sustained during operation.

Before deployment

- Good engineering contexts reduce defects before operation.
- Fault tolerance is used where full failure avoidance is not possible.
- Defensive design reduces avoidable faults.

In operation

- Recovery mechanisms reduce service loss after disruption.
- Maintenance and monitoring sustain trust.
- Organizational processes keep evidence inspectable.

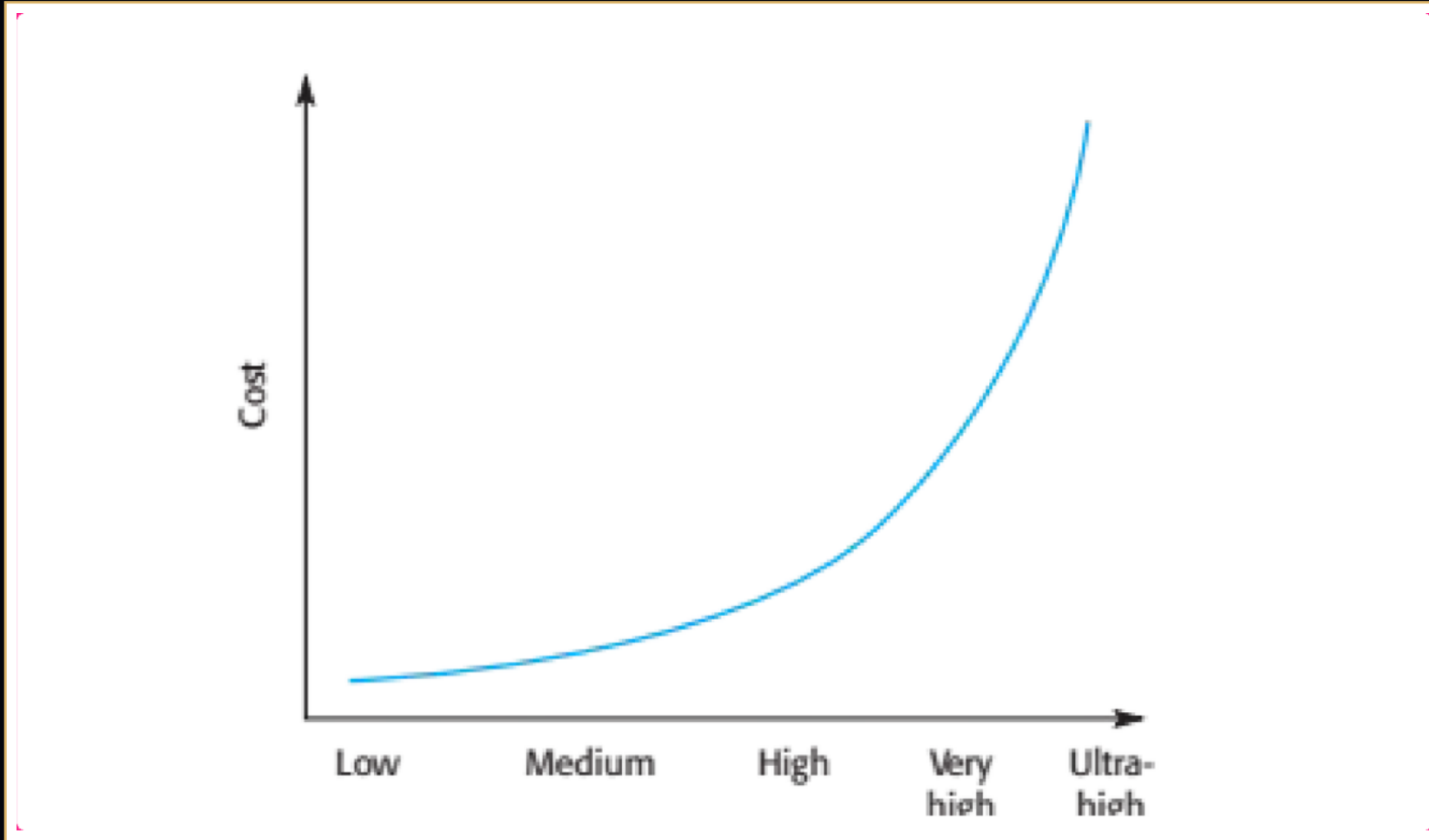
DECISION BRIDGE

Which prevention or recovery action changes your confidence in the system?

TIMEBOX: 60-90 sec

YOUR TASK Name one prevention action and one recovery action for your local system.

Dependability cost trade-off



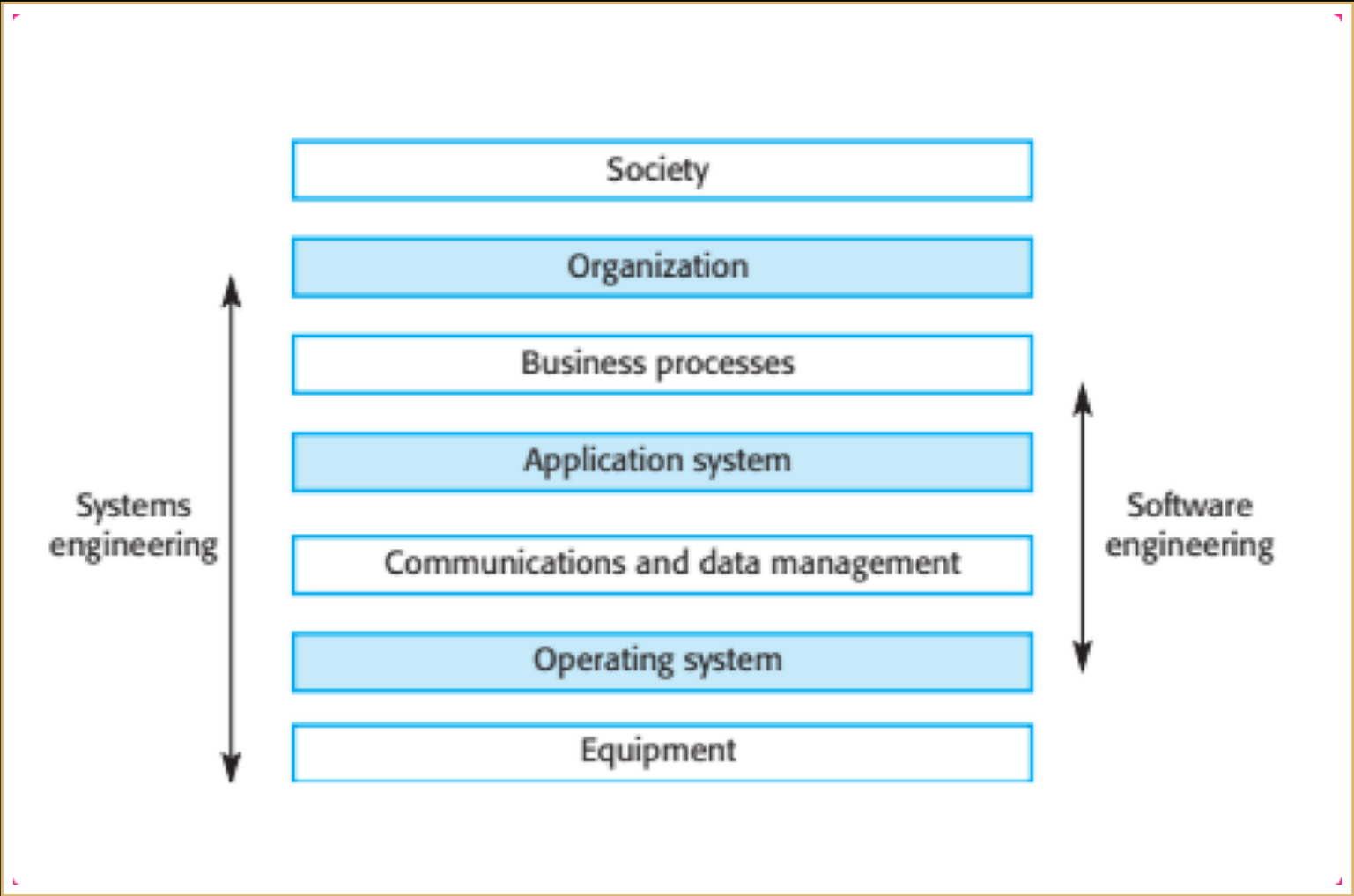
READ THE SOURCE FIGURE

- Assurance cost rises non-linearly as failure risk is pushed down.

- The engineering task is to justify the point where evidence is enough for the risk boundary.

Decision rule: shrink commentary before shrinking the evidence.

Sociotechnical systems - the full stack



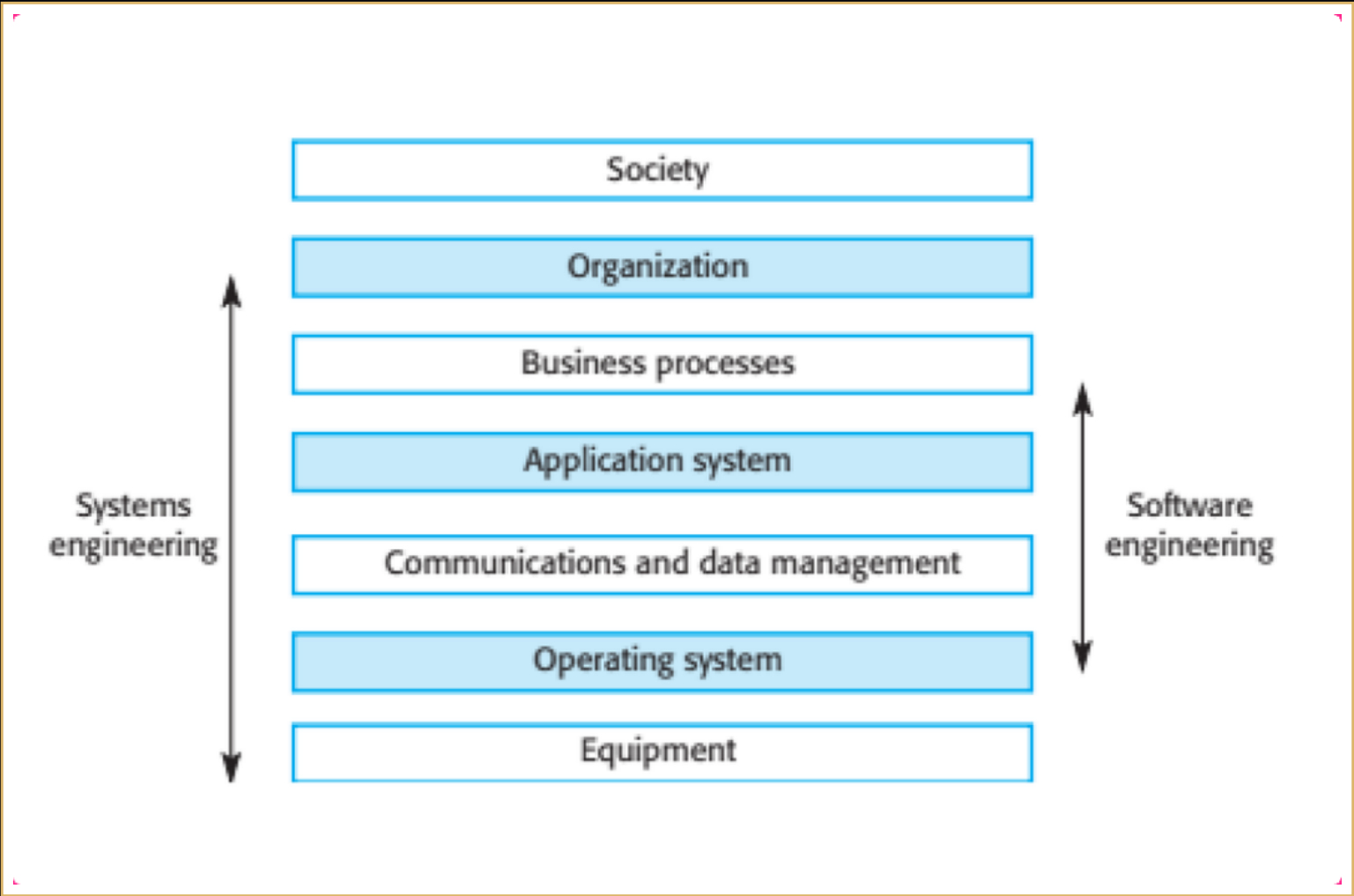
READ THE SOURCE FIGURE

- Dependability is produced by people, process, hardware, software, and operating context together.

- A technical fix can fail if the surrounding workflow contradicts it.

Decision rule: shrink commentary before shrinking the evidence.

Sociotechnical layer as evidence



READ THE SOURCE FIGURE

- Measurement must name which layer changed: staff, workflow, environment, or software.

- Falsification is stronger when the claimed layer is visible.

Decision rule: shrink commentary before shrinking the evidence.



RULE 10 · Known - unknown - monitor

Known, unknown, monitor

Separate evidence from uncertainty before approval.



Approval = evidence state + uncertainty statement + monitoring plan.

TIMEBOX: 60-90 sec

YOUR TASK Separate known facts from unknowns, then monitor the decision-sensitive risk.



Regulation and compliance — evidence is part of the system

Critical-system approval is not a software-only decision.

Why regulation exists

- Governments limit freedom of private companies through standards to keep products safe and secure.
- Many critical systems require approval before entering service.

Whole sociotechnical system

- Compliance applies to the system as a whole, not only the software.
- Certification requires evidence that rules were followed and documentation is auditable.

DECISION BRIDGE

What evidence would an external regulator need before approval?

TIMEBOX: 2 min

YOUR TASK For the Saudi/local case, name the evidence an external regulator would need before approval.

Saudi/local application

Micro-case first, then transfer the same reasoning to the local system.

MICRO-CASE FIRST

A redundant monitor fails during handover because both channels rely on the same maintenance process.

Ask: which source mechanism failed?

Answer path: common-mode failure -> missing independent evidence -> approval boundary.

same logic



TRANSFER TO LOCAL CASE

Saudi hospital medical-device deployment: software passes tests, but maintenance process and organizational sign-off are weak.

Decision question: which STS layer blocks approval?

Mechanism -> Evidence -> Decision boundary

Boundary: micro-case is ISCARB scaffold; local decision uses only mechanisms already taught from P1.

TIMEBOX: 1 min + 5 min

YOUR TASK Solve the micro-case, then transfer the mechanism-evidence-boundary chain to the Saudi case; name the accountable owner.



Redundancy and diversity — useful, but not automatically safe

Independence is the assumption that makes redundancy and diversity valuable.

Redundancy

- Multiple identical or backup components reduce single-point failure.
- Protection depends on failures not occurring at the same time or for the same reason.

Diversity

- Different implementations reduce identical design bugs.
- But diversity increases complexity and may introduce new interactions.

DECISION BRIDGE

Which common-mode failure could defeat the redundancy or diversity choice?

TIMEBOX: 2 min

YOUR TASK

Choose redundancy or diversity for one failure mode; name the common-mode failure that could defeat it.



RULE 12 · Owner - evidence - sign-off

Owner, evidence, sign-off

A dependable answer names who owns the evidence.



If no one owns the evidence, the assurance claim is not inspectable.

TIMEBOX: 2 min

YOUR TASK Name the owner, evidence owner, and sign-off point for the design decision.



DECISION BOX: approve only if process evidence is inspectable - activities, documentation, and V&V trail.

SOURCE SLIDES 31–36

Dependable processes — what “good process” means

Repeatability is necessary, but assurance also requires evidence outsiders can inspect.

Foundation

- A dependable process is well-defined and repeatable.
- Regulators use process information to judge whether good software engineering practice was followed.

Additional attributes

- Auditable: outsiders can understand and check the process.
- Diverse V&V includes redundant and diverse activities.
- Documentable: activities and required documentation are defined.

DECISION BRIDGE

Which process attribute must be visible in an inspectable artifact?

TIMEBOX: 2 min

YOUR TASK Pick two process attributes and explain what evidence would prove they exist in practice.



RULE 13 · Stress variable + fails first

Stress test: change one source variable

Find the first assumption that fails.

1 CHANGE VARIABLE

Certification evidence becomes mandatory.

2 FAILS FIRST

Pure-agile documentation no longer satisfies assurance need.

3 REDESIGN

Add traceability, approval evidence, and named review ownership.

Stress testing is useful only when the first broken assumption is named explicitly.

TIMEBOX: 90 sec

YOUR TASK Change one source variable and state which assumption fails first; add one measurable indicator.



RULE 14 · Practitioner workload

Practitioner workload

Dependability has a human cost as well as a technical cost.

DOCUMENTATION BURDEN

Certification creates process and product evidence that must be produced, reviewed, and maintained.

REVIEW WORKLOAD

Inspections and static analysis consume time, specialist attention, and explicit review roles.

DELIVERY TRADE-OFF

Too little evidence risks approval; too much evidence can slow delivery without improving the decision.

INDUSTRY VARIABLE

Balance documentation burden against delivery speed - then justify the boundary with evidence.

A design that practitioners cannot operate is not dependable in practice.

TIMEBOX: 90 sec

YOUR TASK Name the practitioner workload and explain whether the design reduces or increases it.



DECISION BOX: keep agile only where certification evidence survives - docs, traceability, conflict checks.

SOURCE SLIDES 37-38

Dependable processes and agility — where the tension really is

Agile can be used, but “pure agile” is difficult when certification needs evidence.

Why assurance constrains agility

- Certification requires process and product documentation.
- Up-front requirements analysis helps expose conflicts that may compromise safety or security.
- This conflicts with minimizing documentation.

What still works

- Iterative development, test-first development, and user involvement can still be used.
- Additional planning and documentation make “pure agile” impractical for dependable systems engineering.

DECISION BRIDGE

Which agile practice survives the assurance requirement - and which evidence remains non-negotiable?

TIMEBOX: 2 min

YOUR TASK Choose one agile practice to keep and one assurance activity that must remain non-negotiable.



RULE 15 · AI assist + human sign-off

AI assist, human sign-off

One sentence can be clearer than another diagram.



AI MAY PREPARE EVIDENCE

AI may draft candidate tests, summaries, probes, or code. Its output is evidence preparation - not approval.

AI MUST NOT OWN ASSURANCE

AI cannot certify, approve, or sign off. The accountable human remains responsible for the engineering claim.

HUMAN SIGN-OFF

A named human reviews the evidence, resolves counter-evidence, and decides whether to accept, revise, or reject.

HUMAN-IN-THE-LOOP

If AI surfaces counter-evidence that is unclear, the human pauses the verdict, escalates, and verifies independently.

TIMEBOX: 90 sec

YOUR TASK State where AI may assist, where human sign-off is mandatory, and what must be monitored.



DECISION BOX: use formal methods when proof/model evidence changes the release verdict.

SOURCE SLIDES 39–45

Formal methods — mechanism, benefits, adoption limits

Do not reduce formal methods to “mathematics = safe”; keep source qualifications.

Mechanism + benefit

- Formal specification, specification analysis/proof, transformational development, and program verification.
- Verification-based approaches prove representations equivalent.
- Formal models expose requirements errors, deadlock, and inconsistencies.

Why adoption remains limited

- Problem owners may not understand the formal specification.
- Costs are visible while benefits are harder to estimate.
- Scaling remains difficult and formal specification is not naturally compatible with pure agile development.

DECISION BRIDGE

When does formal evidence materially change the release verdict?

TIMEBOX: 2 min

YOUR TASK Name one error formal methods can expose and one reason an organization may still reject them.



RULE 16 · Portfolio + readiness probe

Build the decision artifact

The artifact must make the trade-off and evidence visible.

SYSTEM

Bound the system and the decision context.

MECHANISM

Name the source-backed mechanism controlling the decision.

TRADE-OFF

State what the option buys, costs, or leaves uncertain.



READINESS CHECK

next test / next monitor

EVIDENCE

Point to an inspectable source, test, calculation, or artifact.

READINESS PROBE

Name the next test or monitor that could change the verdict.

OUTPUT

One inspectable artifact: decision + trade-off + evidence + uncertainty + next test.

TIMEBOX: 5-7 min

YOUR TASK Build one inspectable artifact with decision, trade-off, evidence, uncertainty, and next test.



RULE 17 · Change constraint + peer critique

Change one constraint

A strong decision survives mutation — or explains why it changes.

1 PICK CONSTRAINT

Change exactly one decision-sensitive variable.

2 RE-RUN MECHANISM

Apply the same mechanism under the new condition.

3 RECORD BREAK

State what fails first and why.

4 PEER CRITIQUE

Have a peer challenge the evidence, not the style.

5 REDESIGN

Revise only where the evidence requires it.

A strong decision survives mutation - or explains exactly why it changes.

TIMEBOX: 3 min

YOUR TASK Change one constraint, re-run the mechanism, record what breaks, then redesign.



RULE 18 · Claim - evidence - warrant

Defend the decision

An answer becomes engineering only when the argument is inspectable.

CLAIM

State the bounded claim you defend.

EVIDENCE

Point to the inspectable source, test, calculation, or artifact.

WARRANT

Explain why that evidence supports the claim.

COUNTER-EVIDENCE

Name the strongest result that could defeat the claim.

RESIDUAL UNCERTAINTY

State what remains unverified and how it will be monitored.

Keep counter-evidence separate from uncertainty: they answer different questions.

TIMEBOX: 4 min

YOUR TASK Defend one decision with claim, evidence, warrant, counter-evidence, and residual uncertainty.

Peer-review quick card

Use two questions in class; keep the full 6 x 4 rubric for post-class scoring.

1

Is the evidence independently inspectable?

Could another team verify the claim without trusting your explanation or the AI/tool that prepared it?

2

What would make the claim fall?

Name the variable, counter-example, missing evidence, or failure condition that would reverse the verdict.

Fast peer review tests inspectability and falsifiability before formal scoring.

TIMEBOX: 3 min

YOUR TASK

Exchange artifacts with one peer; answer Q1 and Q2 only.



RULE 20 · Assurance chain + bounded verdict

Take-home decision

Close with a bounded verdict, not a slogan.

APPROVE

Evidence supports the bounded decision.

CONDITIONAL

One named condition must be verified before release.

REDESIGN

The mechanism may be sound, but the current application is not.

REJECT

Evidence contradicts the claim or remains too thin to approve.

EVIDENCE CHAIN

Claim -> Evidence -> Warrant -> Counter-evidence -> Residual uncertainty -> Verdict

TIMEBOX: 3 min

YOUR TASK Deliver the bounded verdict with one P1 anchor, one artifact, counter-evidence, and next verification.



ISCARB · CLOSE

Chapter 10 complete

Dependable Systems

The decision

Can you state a bounded engineering verdict?

The evidence

Can another engineer inspect your source and artifact?

The failure

Can you name what fails first when the constraint changes?

The next step

Can you transfer the logic to a Saudi/local system?

Exit check: if the visual helped but the evidence is missing, the decision is not ready.

TIMEBOX: post-class

YOUR TASK Write your final verdict and the one piece of evidence you still need.